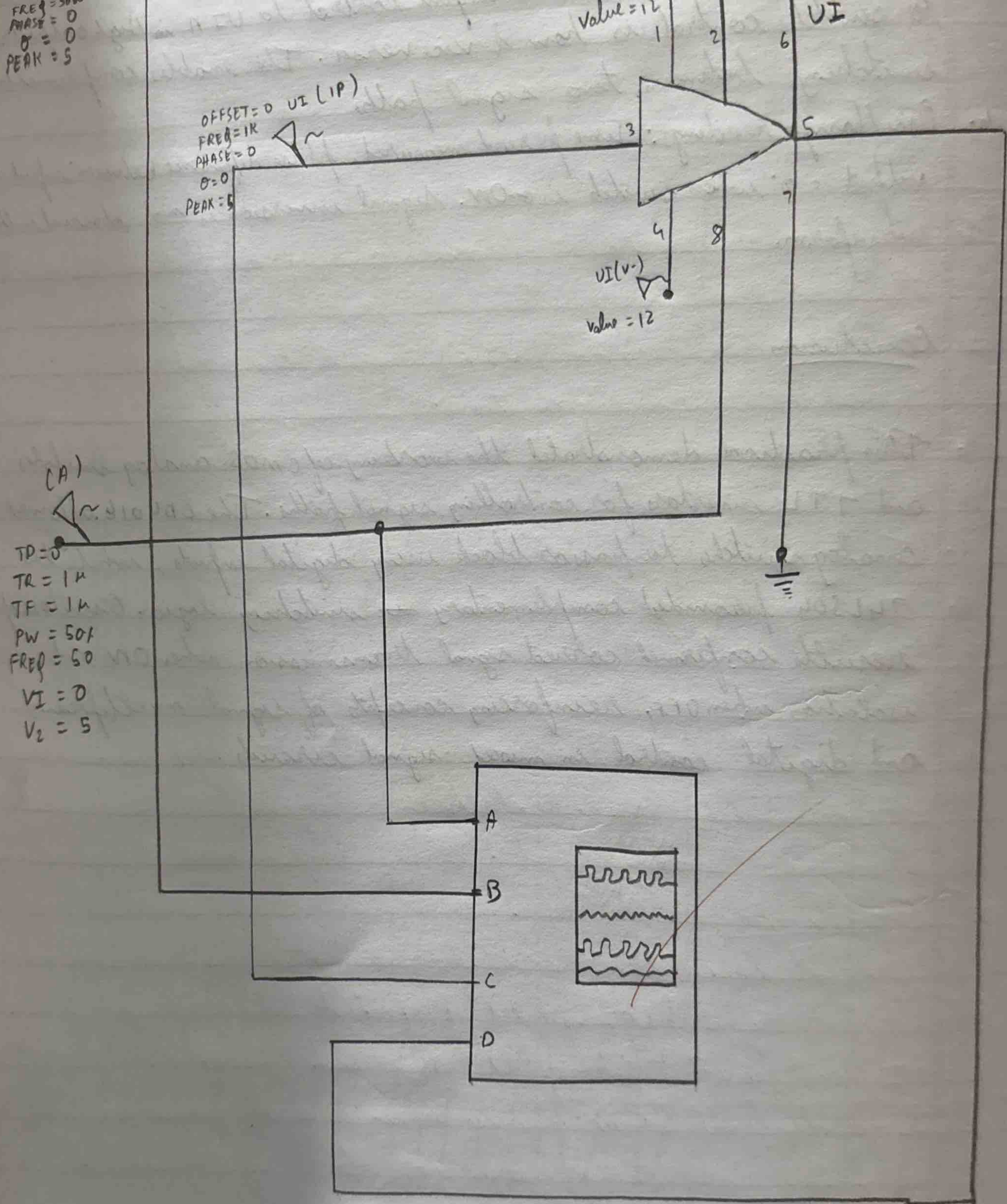


FREQ=
PHASE = 0
θ = 0
PEAK = 5
Value = 12
1
2
6
VI
OFFSET = 0 VI (IP)
FREQ = 1K
PHASE = 0
θ = 0
PEAK = 5
3
5
7
4
8
VI(V-)
Value = 12
(A)
TD = 0
TR = 1μ
TF = 1μ
PW = 50%
FREQ = 50
V1 = 0
V2 = 5
A
B
C
D

Implementation of FSK.

* Frequency Shift keying (FSK)

→ It is a digital modulation technique in which the frequency of the carrier signal is varied according to the digital input data, while the amplitude and phase of the carrier ~~f~~ remain constant

* Binary FSK :-

→ In Binary FSK :-

- Binary 1 is represented by a carrier of higher frequency (f_1).
- Binary 0 is represented by a carrier of lower frequency (f_2).

* Mark frequency:

→ The carrier frequency transmitted when the digital input is logic 1.

* Space Frequency:

→ The carrier frequency transmitted when the digital input is logic 0.

* Modulating signal:

→ A digital pulse signal that controls the switching between two carrier frequencies

* Advantages of FSK :

(i) Better noise immunity than ASK.
(ii) Suitable fore wireless communication.
(iii) Simple demodulation using filters.

* Disadvantages of FSK :

(i) Requires more bandwidth than ASK.
(ii) Lower spectral efficiency compared to PSK.
(iii) Requires accurate frequency contro.

* Application of FSK :

(i) Modems
(ii) Bluetooth communication
(iii) Walkie - talkie
(iv) Wireless telemetry
(v) Caller ID systems)
(vi) Paging system

* Components required:

1.) LF 398N IC
2.) Resistors (47 KΩ)
3.) Two signal generators : Carrier 1: Sine wave (5V, 500 Hz)
: Carrier 2: Sine wave (5V, 1 KHz)

4.) Digital Pulse Generators: Frequency : 50 Hz.
Duty cycle : 50%
5.) CRO (cathode Ray Oscilloscope).
6.) Power supply ($\pm$ 12V)
7.) connecting wires

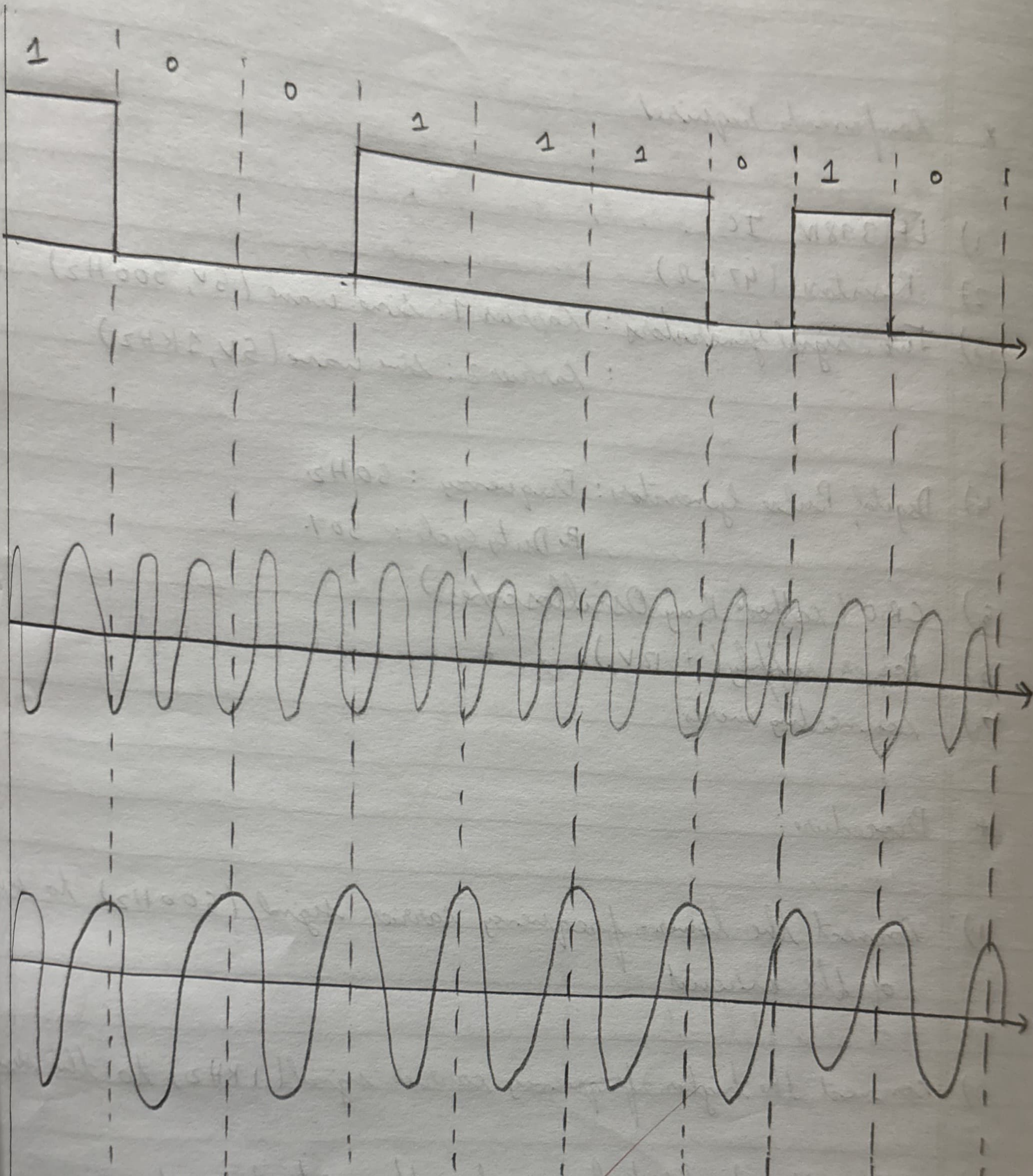
* Procedure:

1.) Connect the lower-frequency carrier signal (500 Hz) to an input of the circuit

2.) connect the higher-frequency carrier signal (1 KHz) to the second input

3.) Apply the digital pulse signal as the control signal.

4.) Connect resistors as per the given circuit diagram.
5.) Power the IC.
6.) Connect the CRO channels.
7.) Switch ON the power supply
8.) Observe the output waveform on the CRO
9.) Verify that frequency changes according to the digital input.


1
0
0
1
1
1
0
1
0
t
t
t

* Observation:

→ when the message signal is High, the output frequency corresponds to f_1 (mark frequency).
→ when the message signal is Low, the output frequency corresponds to f_2
→ The amplitude of the carrier remains constant

* Result.

→ FSK was successfully implemented using two carrier signals of different frequencies. The output waveform shows frequency variation according to the digital input.

* Conclusion:

→ The experimenting verifies that the principle of FSK, where the digital data is transmitted by changing the frequency of the carrier signal. FSK provides better noise immunity than ASK. and is widely used in wireless and data communication systems, although it requires more bandwidth.

24/3/26